A Minor Project-I Report on

BLOODWORKS

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Software Engineering at Pokhara University

By

AAYUSH SHRESTHA

NARAYAN DHAKAL

SUBASH SAPKOTA

Department of Research and Development

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Lamachaur, Kaski, Nepal

(November 2023)

A Minor Project-I Report on

BLOODWORKS

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Software Engineering at Pokhara University

By

AAYUSH SHRESTHA

NARAYAN DHAKAL

SUBASH SAPKOTA

Supervisor

ER. PRATIKSHYA SHRESTHA

Department of Research and Development

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Lamachaur, Kaski, Nepal

(November 2023)

ACKNOWLEDGEMENT

We extend our sincere appreciation to Er. Pratikshya Shrestha, our project supervisor, for her invaluable advice and guidance throughout the project's development. Our sincere gratitude goes to Mr. Birendra Khadka, the principal, and Mr. Ashok Raj Parajuli, the vice principal, for granting us the opportunity to enhance our skills and supporting us in bringing the project to fruition. We also want to express our thanks to the entire supervisory team for their unwavering assistance in resolving the various challenges we encountered during the project.

Special thanks to Er. Rajendra Bahadur Thapa, the Head of the Project Department, for his guidance and support. Additionally, we are thankful to our friends, with whom we have collaboratively discuss about the problems and found solutions over the past few months. Lastly, we appreciate the library staff for their invaluable assistance in providing research materials and valuable suggestions. The entire team is profoundly grateful to all those who contributed to the successful development of BLOODWORKS.

ABSTRACT

Bloodworks is a web application designed to address the challenges associated with blood shortages in the medical field. The project focuses on creating a user-friendly platform for blood donors and blood banks. It provides a means for blood banks to organize and promote their blood donation events, attracting more donors to participate. Individuals in need of blood can utilize Bloodworks to search for potential donors or connect with blood banks directly to their hands on the right type and quantity of blood. By making it easier for people to get access to potential blood donors and enabling direct communication between donors and recipients, Bloodworks aims to solve the problem of blood scarcity and improve how blood donations work in healthcare.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

Chapter 1
INTRODUCTION

1.1 BACKGROUND

Blood donation refers to a practice where people donate their blood to people so it helps them with their health problems. (Blood Donation Essay for Students and Children | 500 Words Essay, 2020). Blood donation is necessary to help provide essential transfusions for patients with various medical conditions and meet the ongoing demand for blood in hospitals and healthcare facilities .

There are two primary methods of donating blood: visiting a healthcare organization directly or participating in donation events. Bloodworks will prioritize the latter approach, focusing on facilitating participation in blood donation events.

Our project aims to offer a comprehensive platform for efficient blood donation event management, specifically designed to boost the number of blood donations. The primary beneficiaries of our initiative are individuals or patients in need of blood. Bloodworks features a user-friendly interface, allowing them to easily search for their required blood type and connecting them seamlessly with potential donors or responsive blood banks that can meet their requirements.

1.2 PROBLEM STATEMENT

Demand of blood is increasing rapidly especially in developing nation. (*14 June World Blood Donor Day*, 2022). In the market today, there are numerous web and mobile applications aimed at partially alleviating this issue. These applications typically offer features like donor registration and the display of donor and blood bank lists. Nevertheless, the fundamental problem often persists unabated. Despite the inclusion of these features, there frequently exists a deficiency in the effective management of donor and blood bank lists. Our site solves this issue by enabling individuals to easily search for their required blood type and find suitable blood banks, all while assisting blood banks in efficiently managing their events and blood inventory.

1.3 OBJECTIVES

- To create a web application for blood management system.

1.4 IMPLICATION

Bloodworks is a web application that presently offers a well maintained list of events and blood inventory for blood banks. The donor lists include a direct contact option, which considerably reduces the time needed to locate the necessary blood. When a requester requires blood, they can simply fill out a request form by clicking on the contact information of a specific donor, which triggers an email notification to inform the donor about the need for blood. This system also aids donors in tracking the number of requests they have received. These features substantially facilitate individuals in obtaining their required blood type, providing invaluable support during times of need.

Chapter 2
LITERATURE REVIEW

There are many applications that we have found that have similar functionalities to Bloodworks. But there are some features we think will differentiate Bloodworks from other sites. Following are some of them:

Nepal Blood:

Nepal Blood is a web-based application created by devLuk Technologies, launched on November 26, 2015. While it shares certain features with Bloodworks, such as donor registration and displaying a list of donors, Nepal Blood lacks the capability for blood banks to add and independently manage their blood donation events (Nepal Blood, 2014).

Hamro LifeBank:

Hamro LifeBank, a web-based application developed by the multi-business company RUMSAN, offers a comprehensive set of features, closely matching those found in Bloodworks. Notably, it includes a unique map feature that Bloodworks does not have. However, it lacks one of the primary features of Bloodworks, which is the display of donor lists (Hamro LifeBank, 2023).

NRCS - Nepal Red Cross Society:

NRCS, a web-based application developed by the International Committee of the Red Cross (ICRC) in 1964, primarily serves as an informational platform for blood donation events and achievements. As a result, it lacks many of the features found on other sites, such as displaying donor lists, managing blood donation events, and more (Nepal Red Cross Society, 2023).

The following is the table comparing the features of all the web applications mentioned earlier including Bloodworks:

Table 2.1: Comparison Table

Web Applications	Display Donor List	Peer-to-Peer Donation	Manage Blood Donation Events	Map	Display News Article
Bloodworks	Yes	Yes	Yes	No	No
Nepal Blood (*Nepal Blood*, 2014)	Yes	Yes	No	No	Yes
Hamro LifeBank (*Hamro LifeBank*, 2023)	No	No	Yes	Yes	Yes
NRCS (*Nepal Red Cross Society*, 2023)	No	No	No	No	Yes

Chapter 3
TOOLS AND METHODOLOGY

3.1 TOOLS

The following tools will be used for the development of the system:

- **HTML, CSS, PHP, JavaScript, and MySQL**: used languages for the coding and implementation of the system.

- **Visual Studio / Sublime Text**: used as the main code editor software.

- **Google Chrome**: used as a browser for viewing output and as a console.

- **XAMPP Server**: used to install the local server for PHP.

- **DIA**: used for illustrating the UML diagrams.

- **GitHub**: used for version control and project management.

- **Online Gantt**: used for creating the time chart table.

- **Canva**: used for assets and images.

- **Font-awesome**: used for icon assets for the website.

3.2 METHODLOGY

The development model that our project will be using is a combination of waterfall and iterative-incremental models. The waterfall model is a linear, sequential approach to the software development life cycle (SDLC) that is popular in software engineering and product development. The waterfall model uses a logical progression of SDLC steps for a project, similar to the direction water flows over the edge of a cliff. It sets distinct endpoints or goals for each phase of development. Those endpoints or goals can't be revisited after their completion. (Lutkevich, 2006).

The procedure we employed included most of the phases of the waterfall model, such as requirement analysis, design, coding, implementation, and testing. However, we incorporated elements of an iterative-incremental model because we believed that our initial project concept was preliminary, allowing room for potential concept changes in the future. This approach enabled us to seamlessly accommodate new features or explore alternative techniques for implementing existing features without encountering developmental challenges.

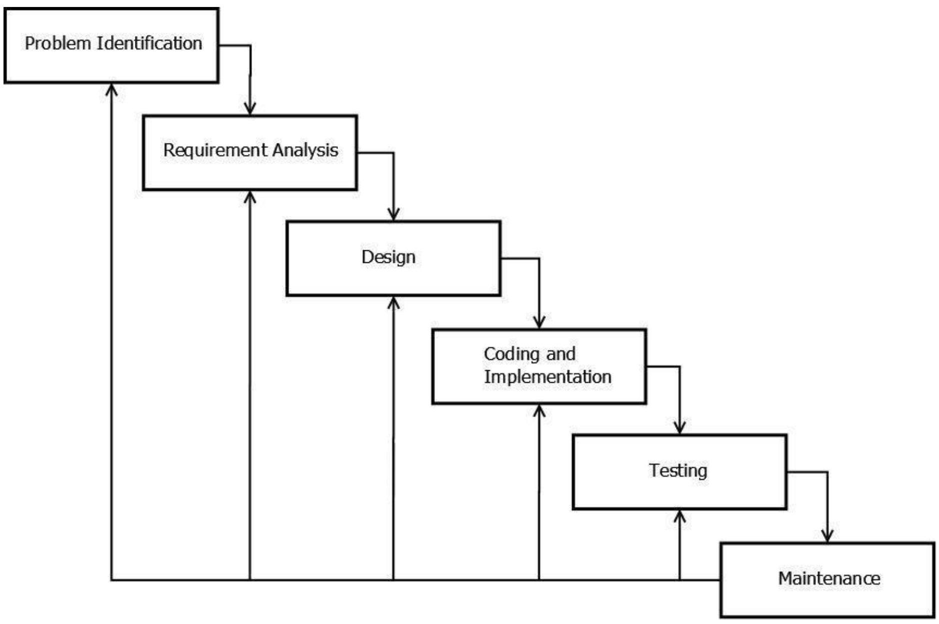


Figure 3.2.1 : Block Diagram of Iterative Waterfall Model

3.3 DESIGN

3.3.1 USE CASE DIAGRAM

The system consists of two primary actors: User and Blood Bank , and below given figure shows the use case diagram concerning them:

Figure 3.3.1.1 : Use Case Diagram for Bloodworks

Use Case UC 1 : Enable as Donor

Primary Actor : User

Pre Condition : User must be logged in.

Post Condition : System shows the approved message.

Basic Flow :

- System shows the Enable as Donor button.
- User clicks the button and system verifies the user.
- System adds the user to the donor list.

Alternative Flow :

- If wrong data are entered then shows an error message.

Use Case UC 2 : Request Blood

Primary Actor : User

Pre Condition : User must be logged in.

Post Condition : System shows the approved message.

Basic Flow :

- System show the request form.
- User fills the form according to the type of blood they want and submit.
- System adds the request to requests list.
- System sends email to potential donors.

Alternative Flow :

- If wrong data are entered then email won't be send and shows an error message.

Use Case UC 3 : Request for Blood Bank Registration

Primary Actor : Blood Bank

Pre Condition : Blood Bank must be an authorized institution.

Post Condition : Blood bank is added and displayed to the list of blood banks.

Basic Flow :

- System shows Add Blood Bank button.
- Blood Bank clicks the button and form is displayed.
- Blood Bank fills the form and submit.

- System admin verifies the Blood Bank Registration.
- System adds blood bank to the Blood Bank List.

Alternative Flow :

- If blood bank is not authorized then system admin mark the blood bank as not verified in the database and it won't be displayed at the give list of blood bank.

Use Case UC 4 : Add/Edit Blood Donation Event

Primary Actor : Blood Bank

Pre Condition : Blood Bank must be an authorized institution.

Post Condition : New Event is created, or existing is updated. System shows the approved message.

Basic Flow :

- System shows Add Event button.
- Blood Bank clicks the button and form is displayed.
- Blood Bank fills the form and submit.
- System admin verifies the Blood Bank.
- System adds event to the Event List and shows Edit Event button.
- Blood Bank clicks the edit button if they want to change any details about the event and edits the detail.
- System updates the Event List.

Alternative Flow :

- If blood bank is not verified then system shows error message.

3.3.2 ENTITY RELATIONSHIP DIAGRAM

An Entity Relationship Diagram is a diagram that represents relationships among entities in a database. An ER Diagram in DBMS plays a crucial role in designing the database(*ER Diagrams in DBMS: Entity Relationship Diagram Model*, 2023). The figure below shows the structure of the Bloodworks's database:

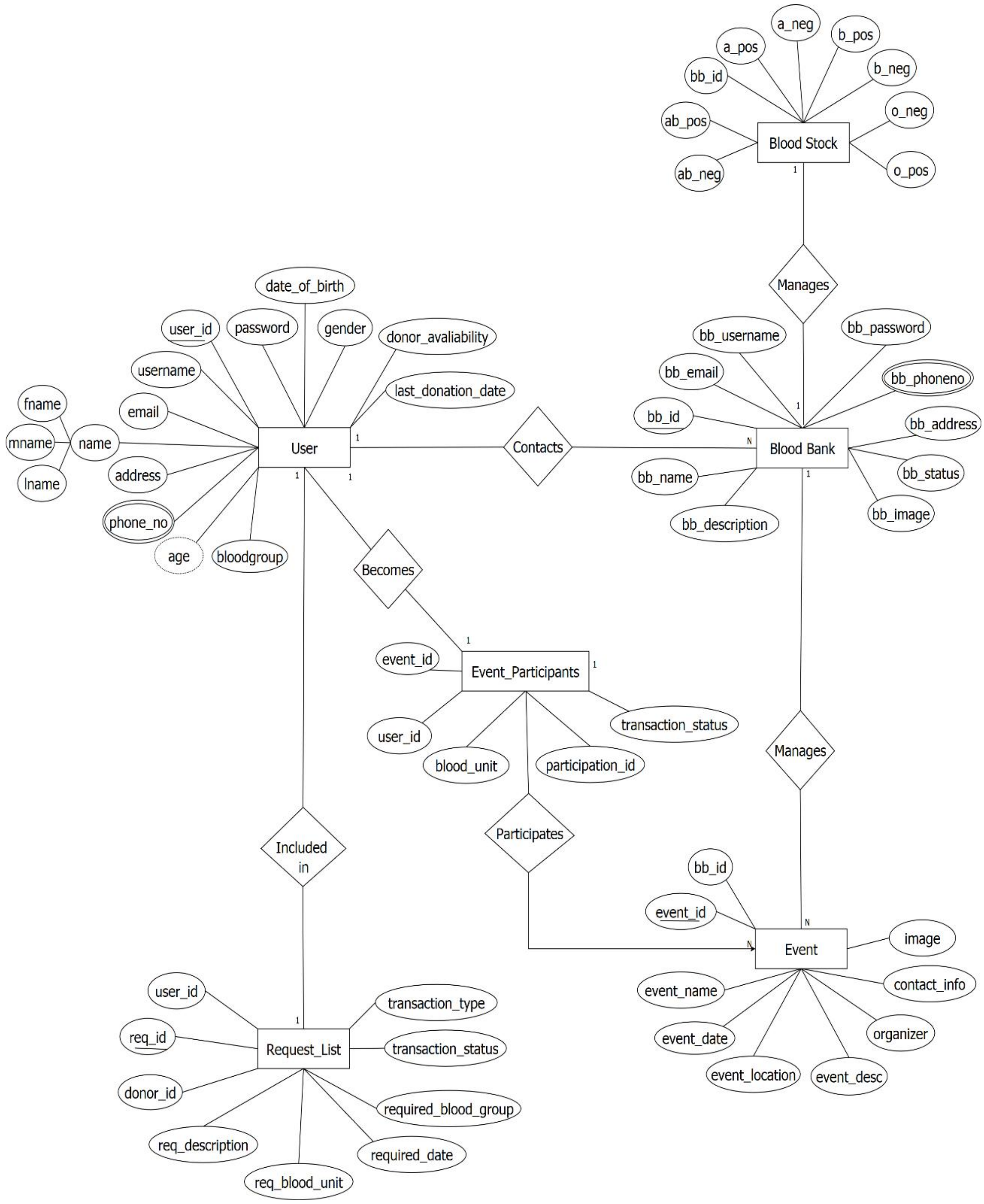


Figure 3.3.2.1 : Entity Relationship Diagram of Bloodworks

3.3.3 SYSTEM SEQUENCE DIAGRAM

The following system sequence diagram shows the scenarios for the operation of enabling user as a donor, requesting blood, blood bank registration and adding/editing blood donation event:

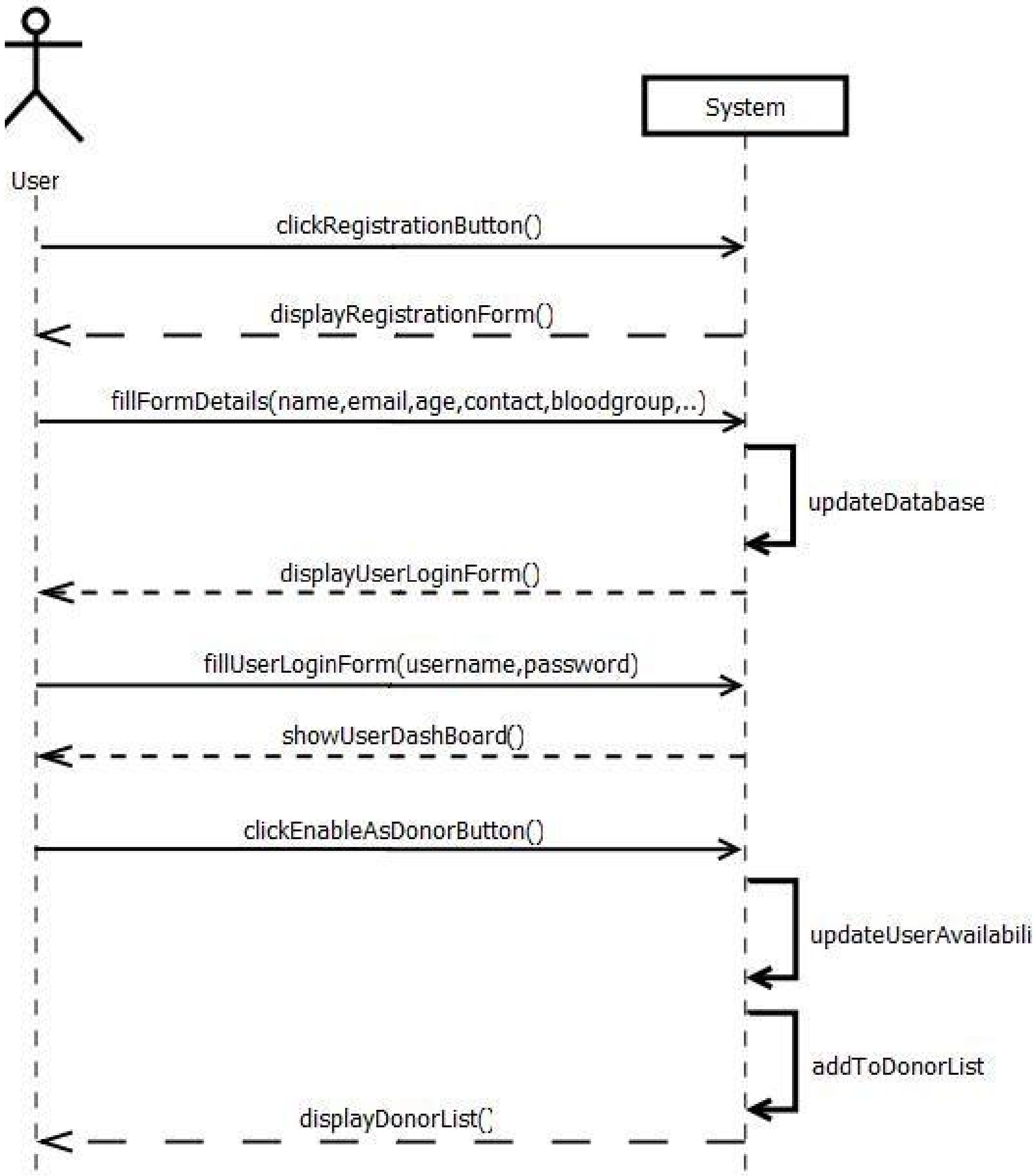


Figure 3.3.3.1: System Sequence Diagram for Enable as Donor Operation

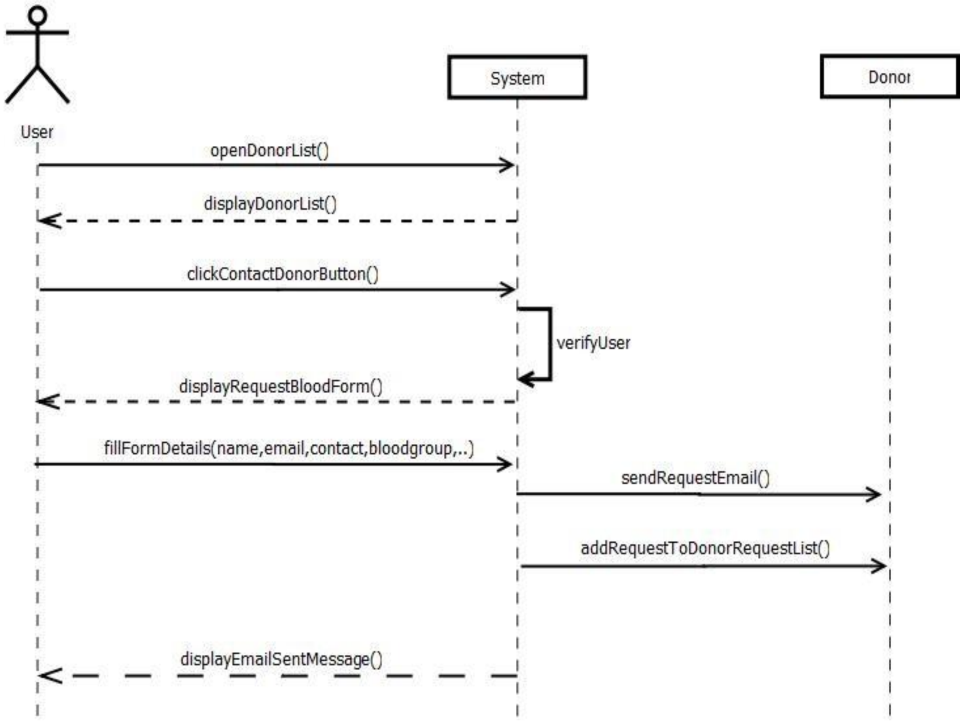


Figure 3.3.3.2: System Sequence Diagram for Request Blood Operation

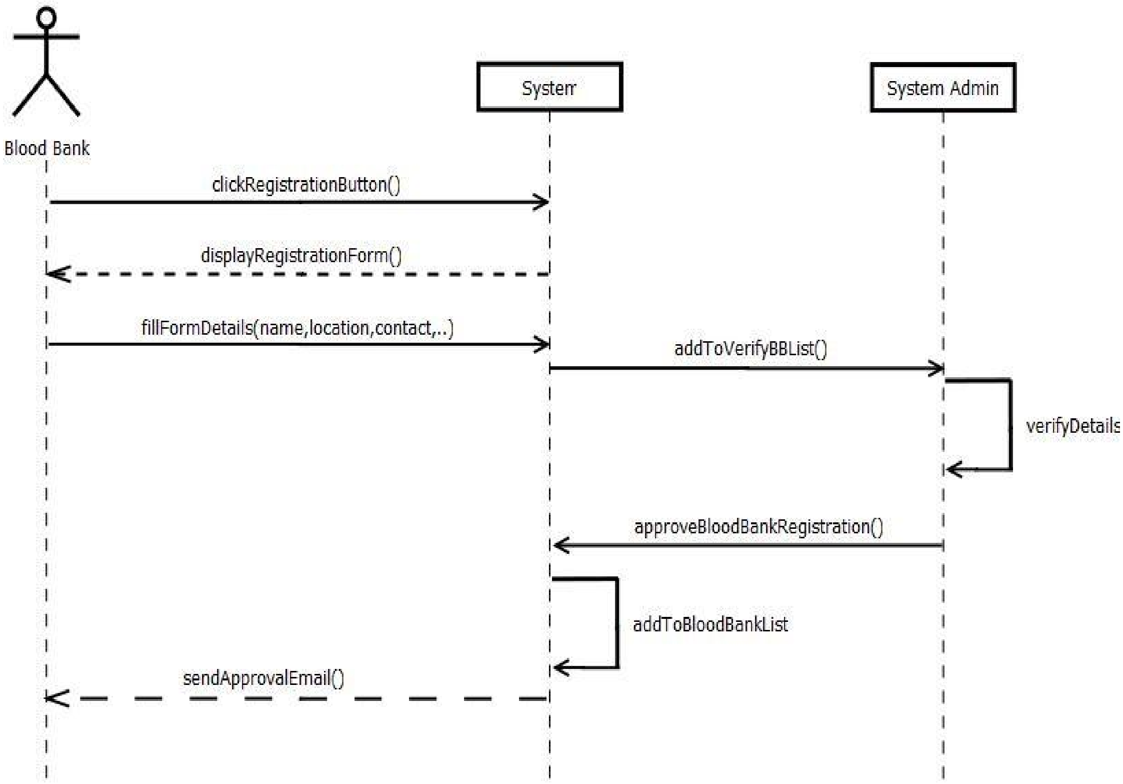


Figure 3.3.3.3: System Sequence Diagram for Blood Bank Registration Operation

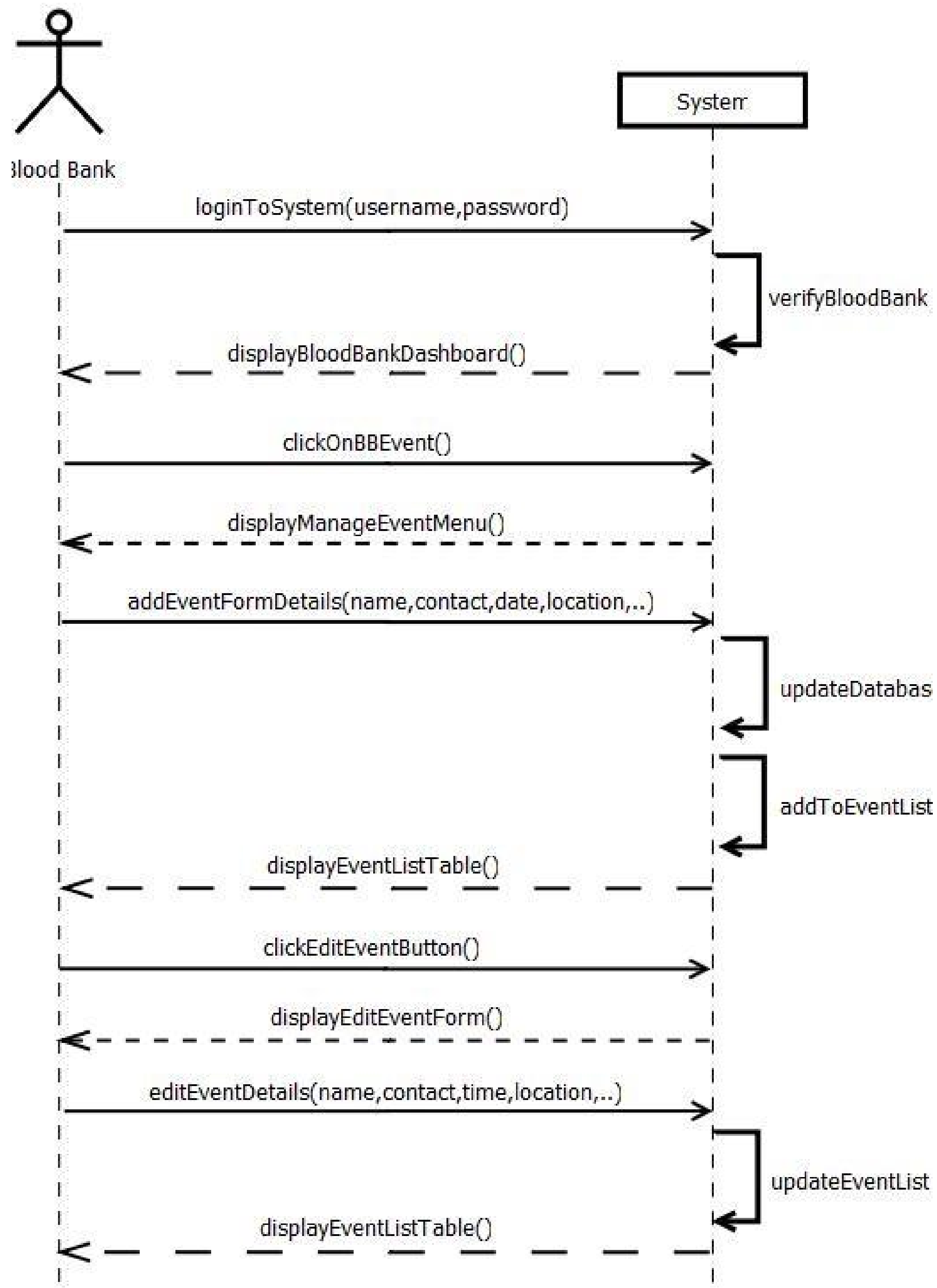


Figure 3.3.3.4: System Sequence Diagram for Add/Edit Blood Donation Event

3.3.4 DESIGN CLASS DIAGRAM

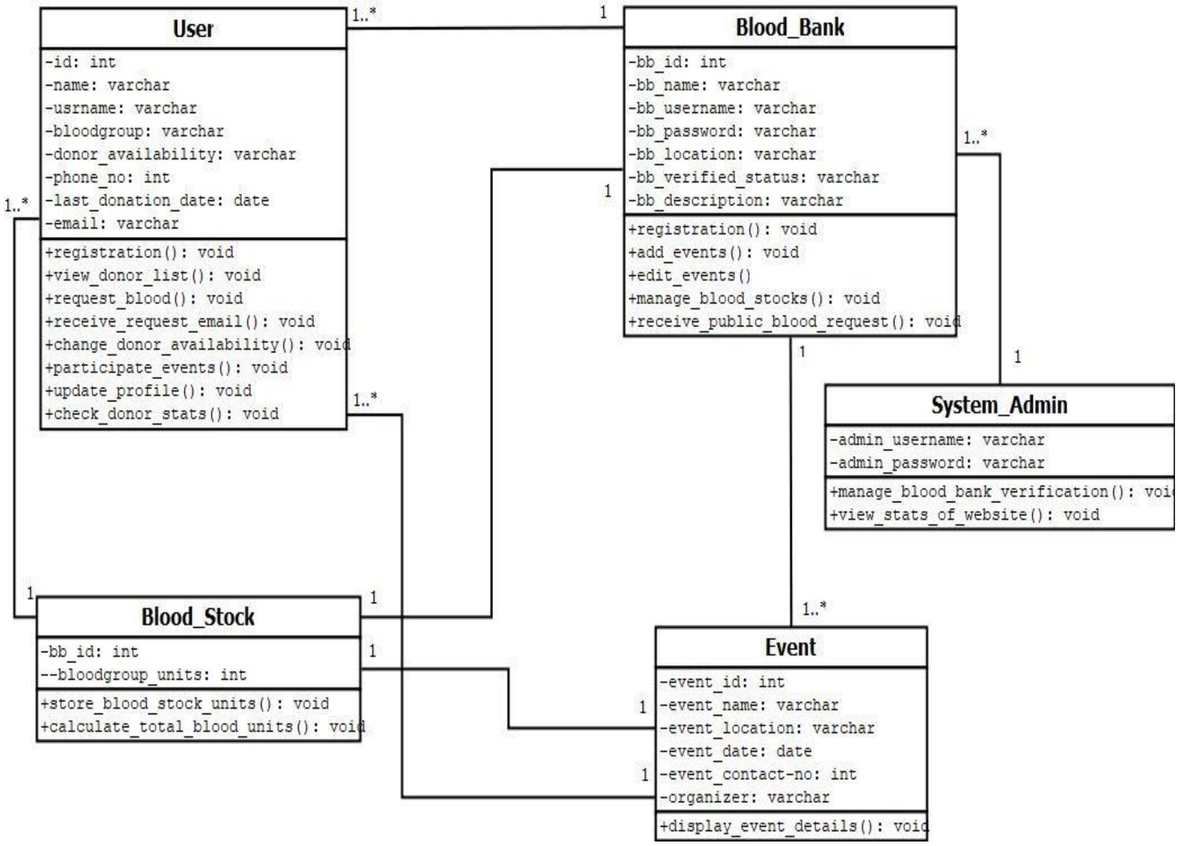


Figure 3.3.4.1: Design Class Diagram of Bloodworks

3.3.5 INTERACTION DIAGRAM

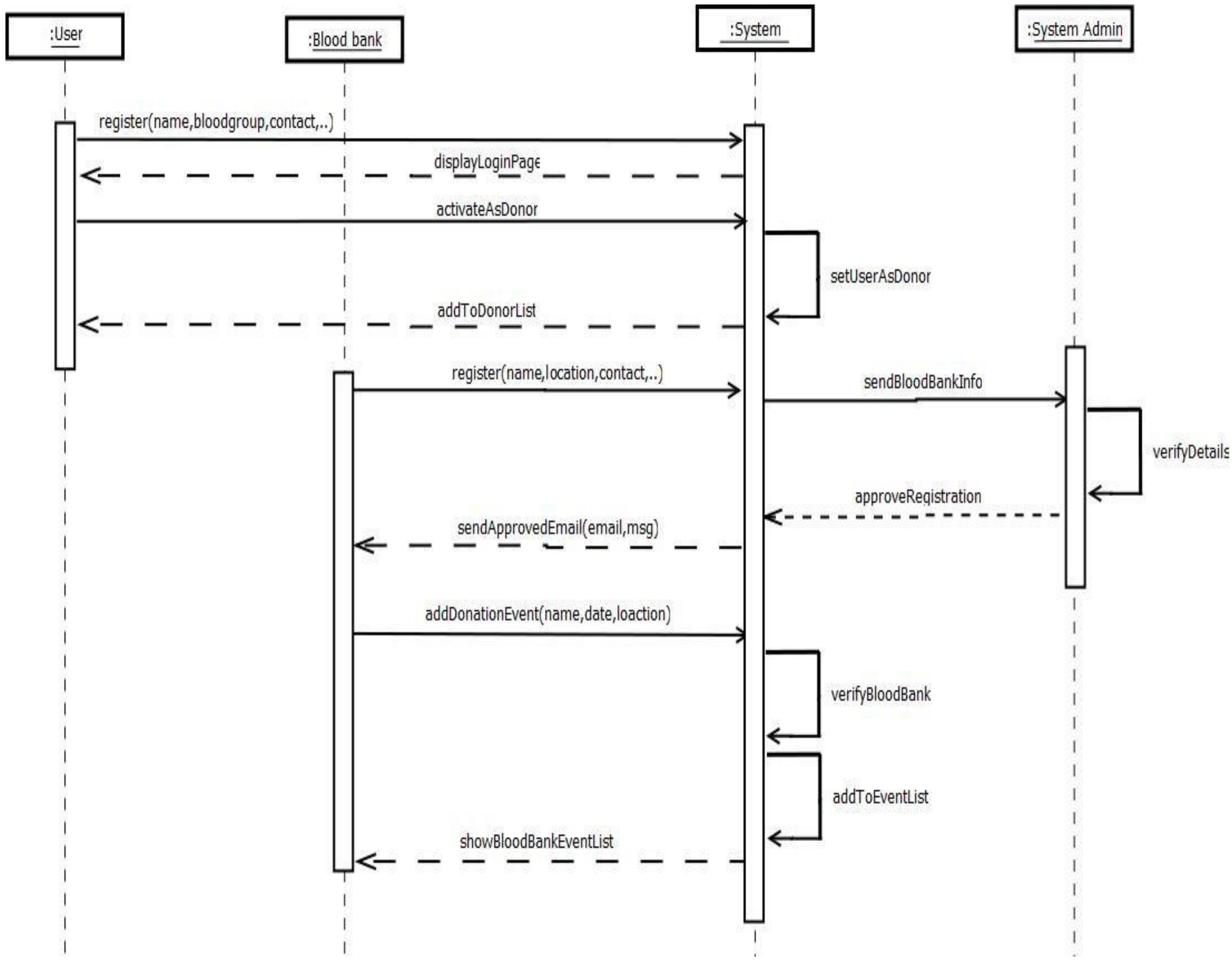


Figure 3.3.5.1: Interaction Diagram of Bloodworks

Chapter 4
TEST CASES

4.1 SOFTWARE TESTING

Software testing is a process to evaluate the functionality of a software application with an intent to find whether the developed software met the specified requirements or not. It also identifies the defects to ensure that the product is defect-free in order to produce a quality product.

4.2 TEST OBJECTIVES

The main objectives of testing BLOODWORKS were to:

- Check whether the web application is as per the requirements or not.
- Ensure defects get a fix from the developers before deployment.

4.3 TEST RESULTS

Table 4.3.1: Test Cases

SN	TEST CASES	EXPECTED	OBSERVED	RESULT
1	Blood Bank Registration Approval	The registration request should shown in the blood bank request table for the admin to approve or decline.	The registration request is being shown in the blood bank request table for the admin to approve or decline.	Ok
2	Blood Request Email	An email should be sent by the system and receive by the specified donor to whom blood is requested.	An email is sent by the system to the specified donor after filling the blood request form by the requester.	Ok
3	Enable as Donor	When a user activates the "Become a Donor" button in their dashboard's settings section, they should be added to the donor list, and removed when deactivated.	To show if a user is available as a donor , the “Become a Donor” switch is activated, and deactivated for unavailability.	Ok
4	Manage Events	Blood banks should be able to add new events and to edit or update their existing events directly from their dashboard settings.	Blood banks can add new events and to edit or update their existing events directly from their dashboard settings.	Ok

Chapter 5
CONCLUSION AND RECOMMENDATION

5.1 CONCLUSION

In summary, our project, Bloodworks is a web-application dedicated to properly managing blood resources by providing a user-friendly platform for efficient blood donation event management and connecting donors with recipients. The website serves as a bridge between recipients, donors, and blood banks, facilitating the easier exchange of necessary blood donations.

5.2 RECOMMENDATION

Bloodworks still isn’t a perfect/complete site, there are still room for improvements. Here are some limitations or features that could be considered for future enhancements:

- Implementation to examine user medical records for eligibility during the registration process.
- Add location tracking and map feature.

BIBILOGRAPHY

Blood Donation Essay for Students and Children | 500 Words Essay. (2020). Toppr. Retrieved May 24, 2023, from https://www.toppr.com/guides/essays/blooddonation-essay/

ER Diagrams in DBMS: Entity Relationship Diagram Model. (2023, May 23). Simplilearn. Retrieved May 27, 2023, from https://www.simplilearn.com/tutorials/sql-tutorial/erdiagram-in-dbms

14 June World Blood Donor Day. (2022, June 14). World Health Organization (WHO). Retrieved May 24, 2023, from https://www.who.int/campaigns/world-blood-donorday

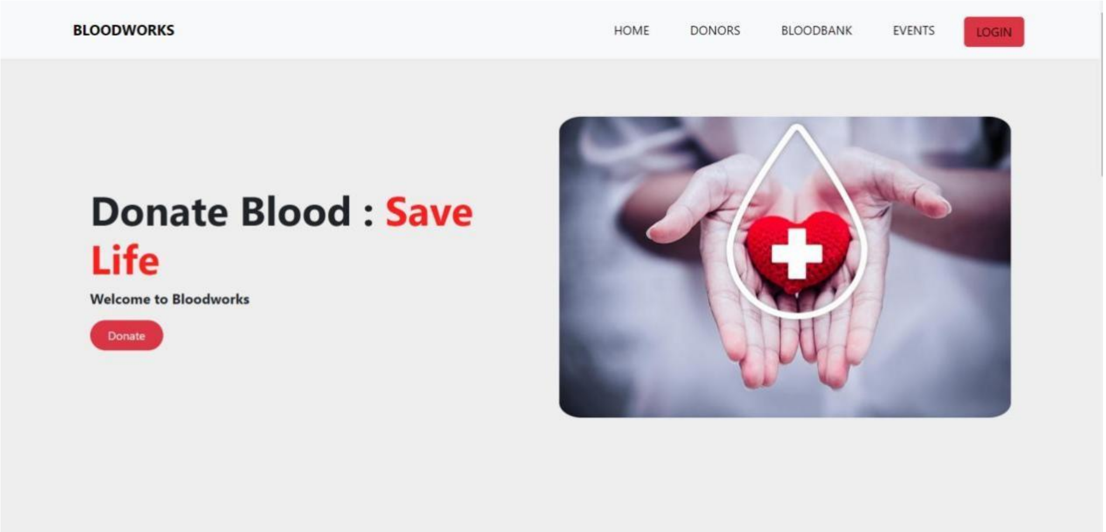
Hamro LifeBank. (2023). Hamro LifeBank - Digital Blood system. Retrieved May 24, 2023, from https://hamrolifebank.com/

Lutkevich, B. (2006). What is the Waterfall Model? - Definition and Guide. TechTarget. Retrieved May 24, 2023, from https://www.techtarget.com/searchsoftwarequality/definition/waterfall-model

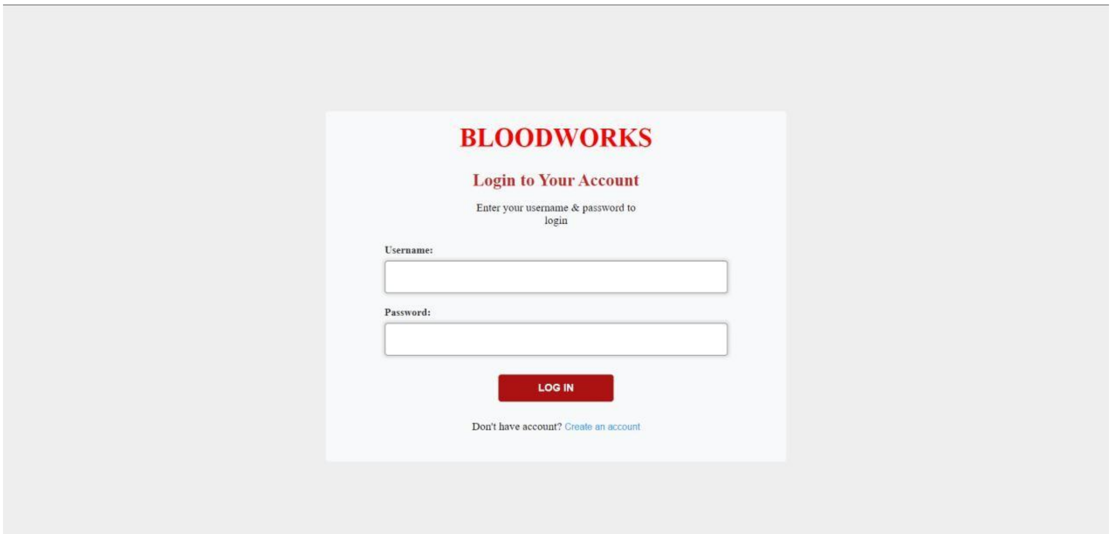
Nepal Blood. (2014). Nepal Blood | Blood Bank Kathmandu | Donate Blood | Fresh Blood. Retrieved May 24, 2023, from https://nepalblood.com/

Nepal Red Cross Society. (2023). Nepal Red Cross Society: NRCS. Retrieved May 24, 2023, from https://nrcs.org/

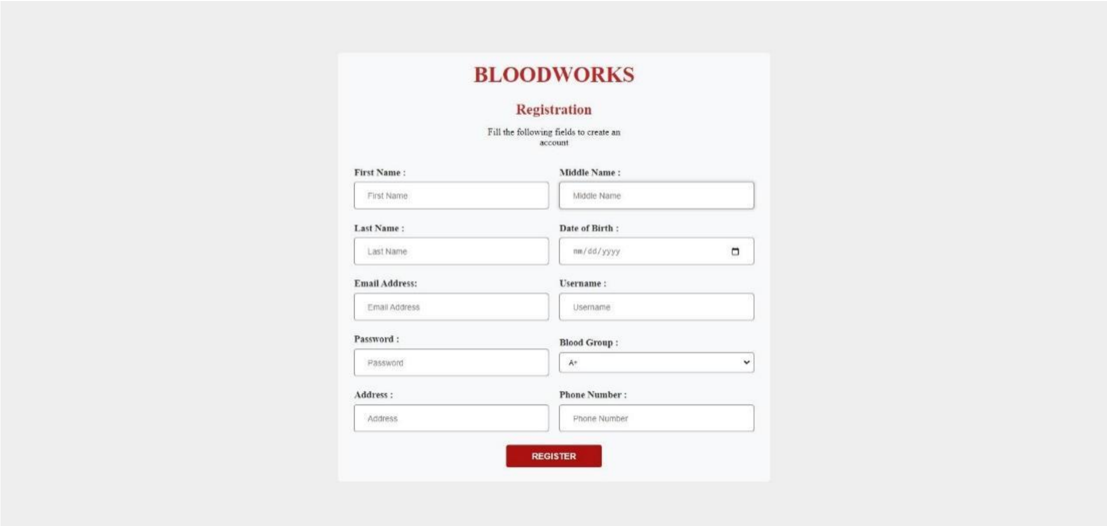
APPENDICES



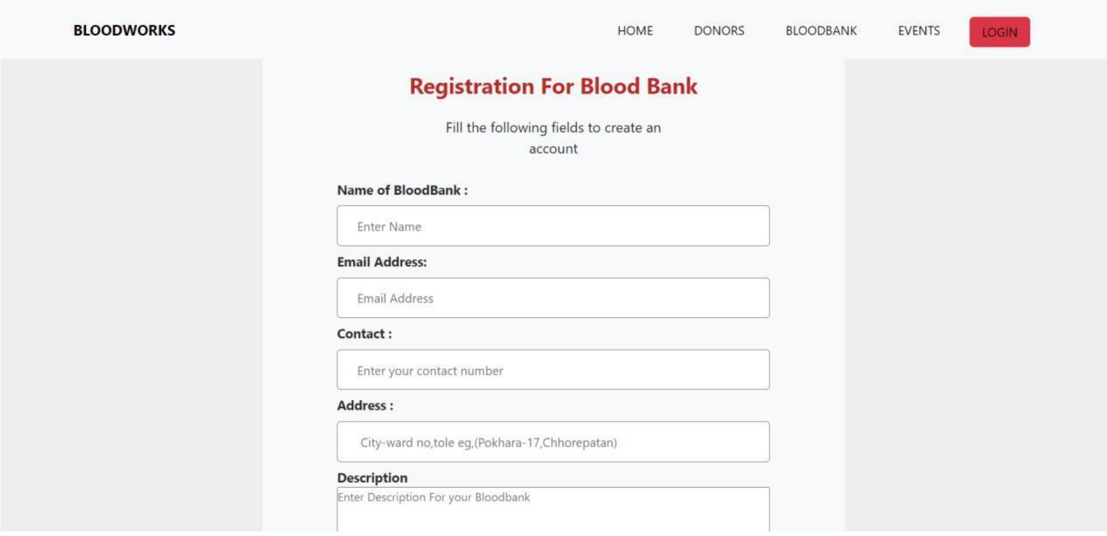
Appendix 1 : Home Page



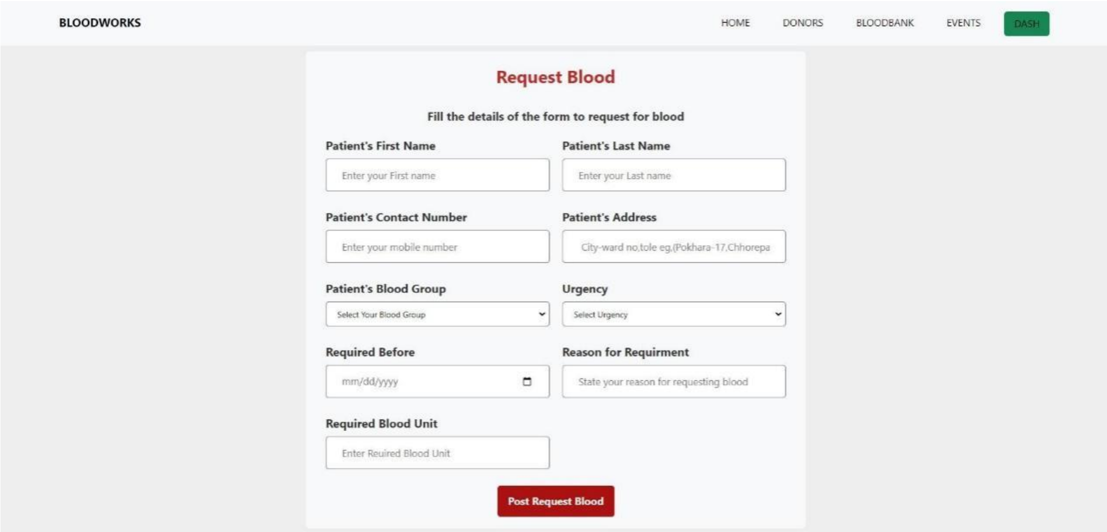
Appendix 2 : Login page



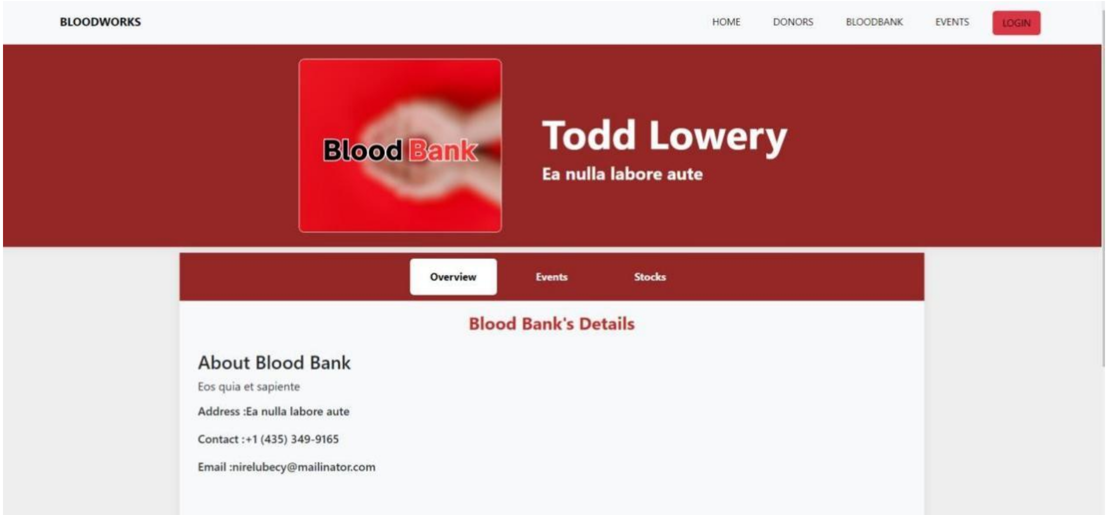
Appendix 3 : Registration Page



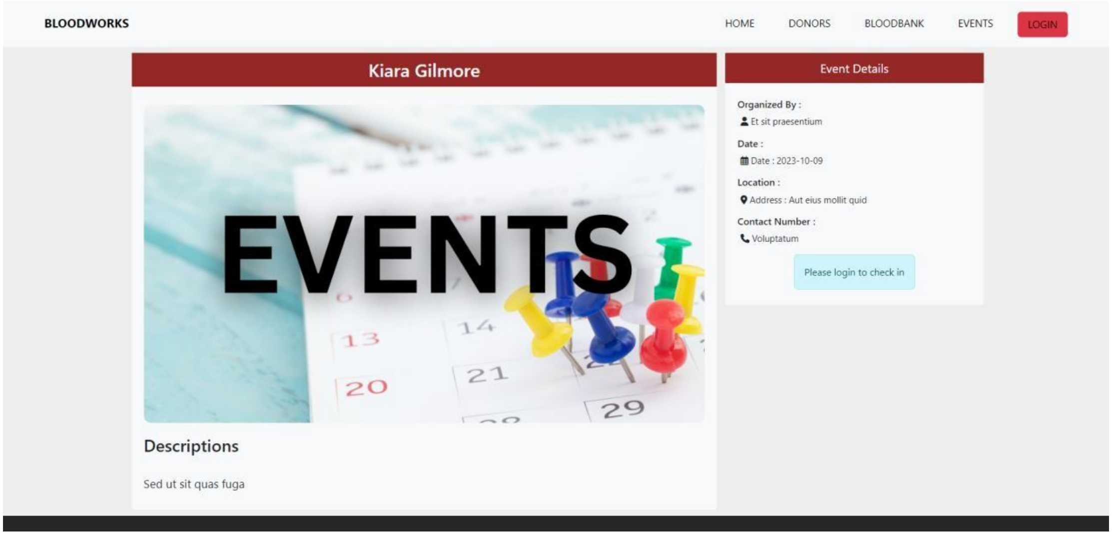
Appendix 4 : Blood Bank Registration Page



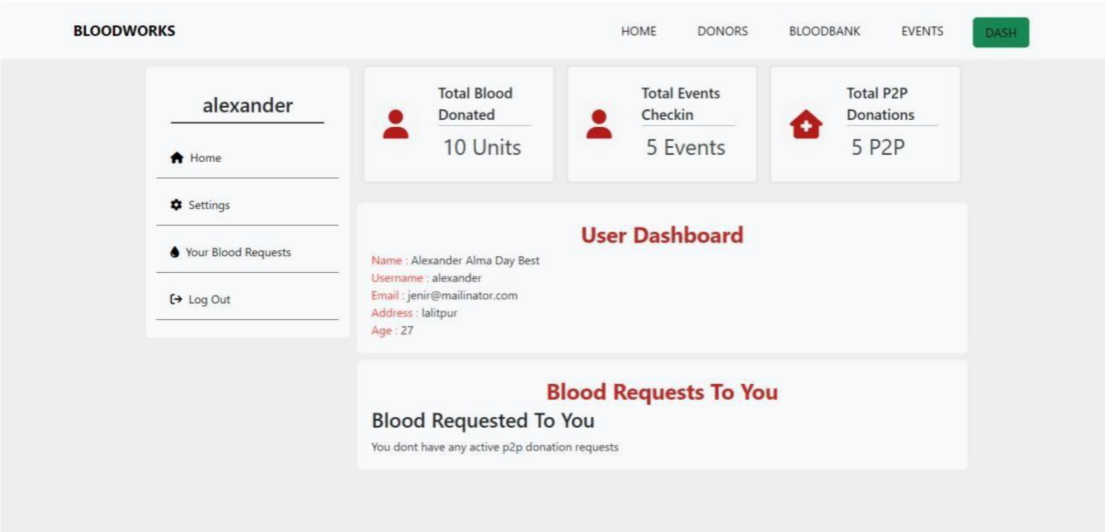
Appendix 5 : Blood Request Form



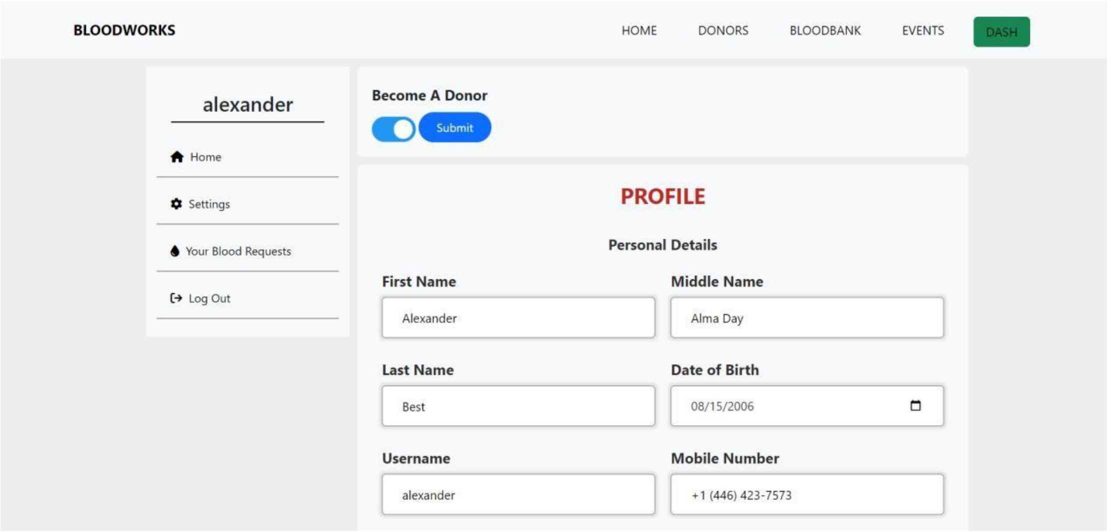
Appendix 6 : Blood Bank Profile Page



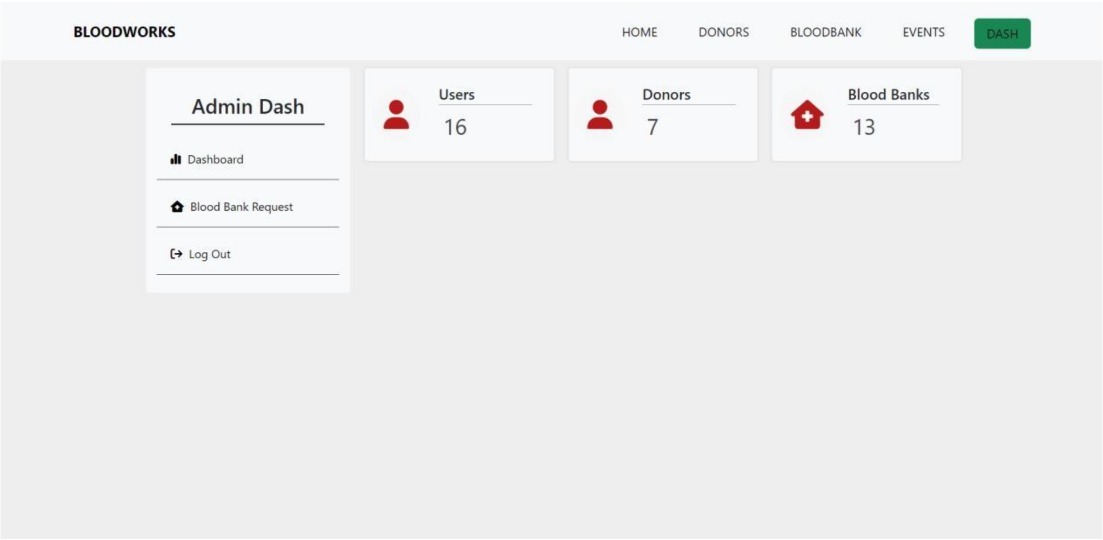
Appendix 7 : Event Profile Page



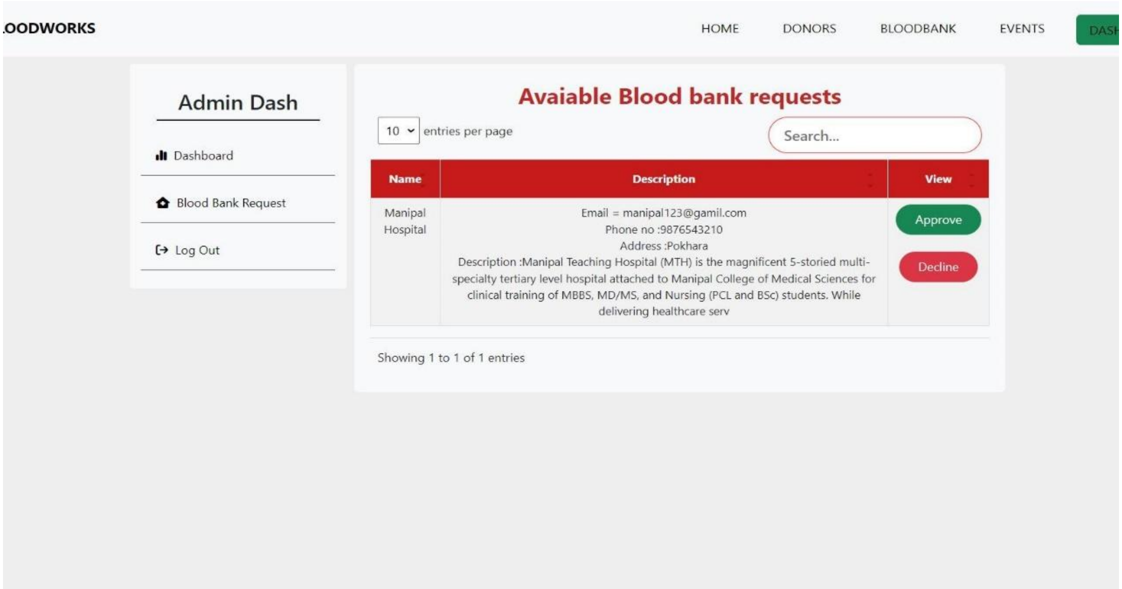
Appendix 8 : User Dash Board



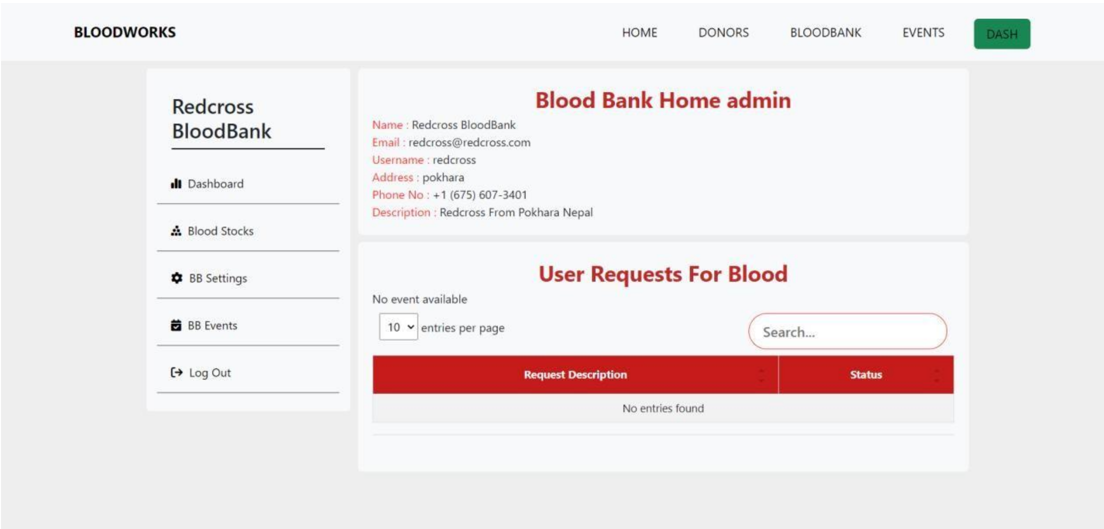
Appendix 9 : User Dash Board Settings



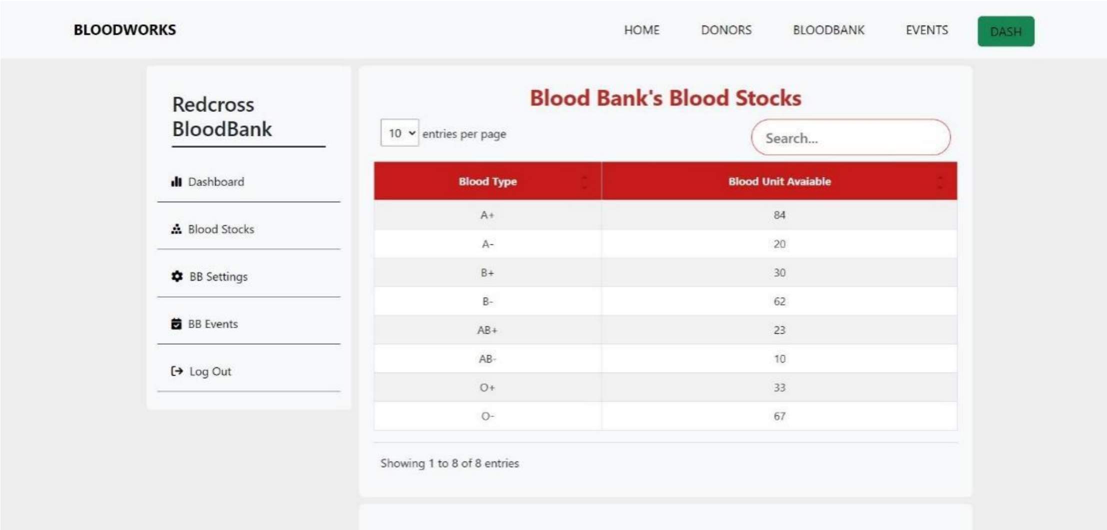
Appendix 10 : Admin Dash Board



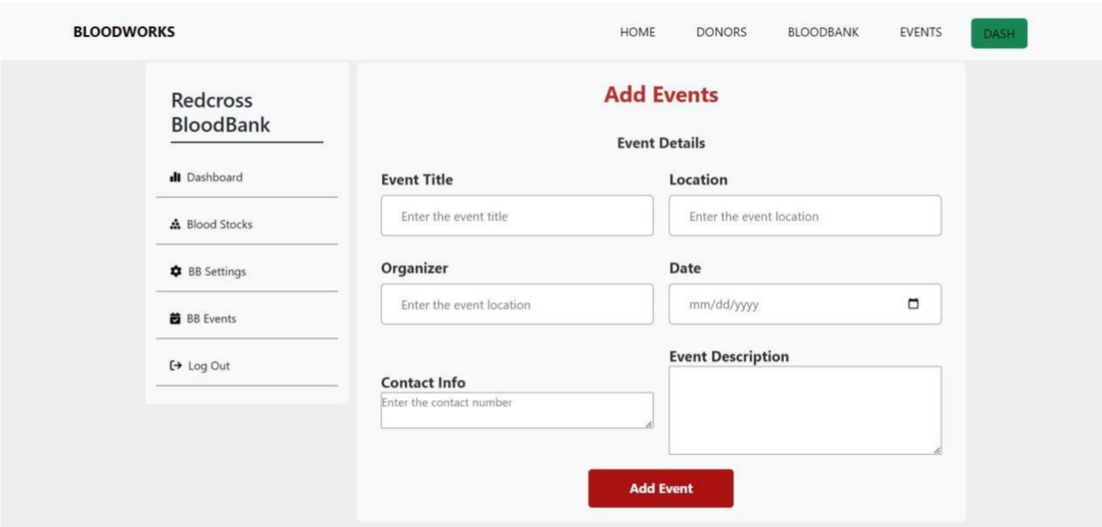
Appendix 11 : Admin Dash Board Blood Bank Registration Request Approval List



Appendix 12 : Blood Bank Dash Board



Appendix 13 : Blood Bank's Blood Stock List



Appendix 14 : Blood Bank's Add/Edit Events Page